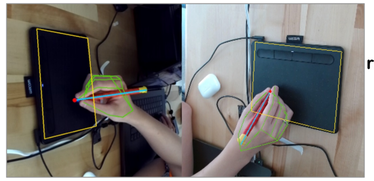
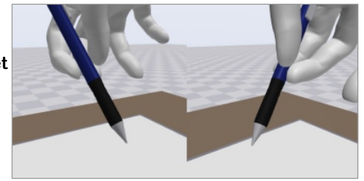
1 Grasp: learn the contact & holding



Human grasp (MediaPipe)

retarget



Grasp mapped to sim hand

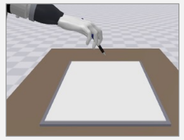
grasp demos

BC init

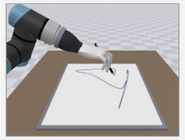
Grasp MLP

PPO

MuJoCo: UR5e + hand + free pen



static / moving hold



env-driven scribble

2 All-task writing BC: learn the writing

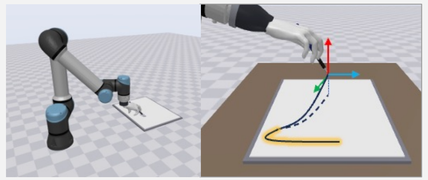
IK teacher

Arm

Grasp MLP
from ①

Fingers

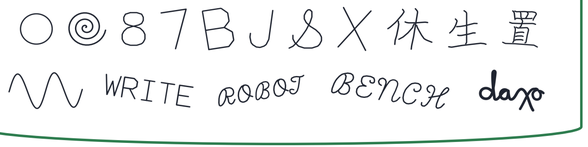
cursor window + state (pen, tip, joints)



teacher rollout

env cursor [x,y,z,f,t]

IK demo set

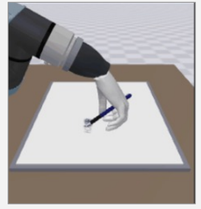


weighted BC

Writing MLP
(arm + wrist)

+ stop signal

eval





BC clone ink

3 Full-task RL on the composed policy

Composed policy π(a | x)

$x_t \in \mathbb{R}^{217}$

137 state + 16 × 5 cursor

Grasp MLP
25 fingers

Writing MLP
7 arm, 2 wrist, stop

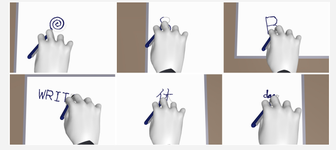
$a_t \in \mathbb{R}^{35}$

obs x_{t+1}

action a_t

PPO + KL step control

Closed-loop rollouts(all tasks mixed)



UR5e + hand + free pen
25 Hz, pen held by friction only

Outcome reward r_t

+ ordered coverage
– off-stroke ink
– force error
– time
– pen drop
– unfinish